

Name: _____ **Date:** _____

Problem 1. Use the squeeze theorem to find the following limit: $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$

Problem 2. Use the squeeze theorem to find the following limit: $\lim_{x \rightarrow 0} x \sin(x)$

Problem 3. Use the squeeze theorem to find the following limit: $\lim_{x \rightarrow 0} x^3 \cos\left(\frac{1}{x}\right)$

Problem 4. Use the squeeze theorem to find the following limit: $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x}$

Problem 5. Use the squeeze theorem to find the following limit: $\lim_{x \rightarrow 0} x^3 \cos\left(\frac{x+1}{x^3 + 3x^2 - 5x}\right)$

Problem 6. Use the squeeze theorem to find the following limit: $\lim_{x \rightarrow 0} xe^{\sin(\frac{1}{x})}$